NPC — Unit 4 Assignment

Configure DHCP and Set Up a Network Using Cisco Packet Tracer

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1. Objective

To configure a DHCP server in Cisco Packet Tracer so that PCs on a LAN automatically receive an IP address, subnet mask, default gateway and DNS server address, and to verify connectivity between the PCs.

2. Topology and IP Plan

PC0, PC1, PC2 and the DHCP server connect to Switch0 (2960). Network: 192.168.10.0/24.

Device	Addressing	IP address
DHCP-Server	Static	192.168.10.2 / 255.255.255.0
PC0, PC1, PC2	DHCP	from pool, starting 192.168.10.10

DHCP pool setting	Value
Pool name	NPC-Pool
Default gateway	192.168.10.1
DNS server	192.168.10.2
Start IP address	192.168.10.10
Subnet mask	255.255.255.0
Maximum number of users	50

Note: no router is in this topology, so 192.168.10.1 is only the gateway address handed out to clients. Add a router with that address if routing is required.



3. Configuration Steps

Step 1 — Build the topology

1. Add 1 switch, 3 PCs and 1 Server. Connect with Copper Straight-Through cables.
2. Rename the server to DHCP-Server and wait for green link lights.

Step 2 — Static IP on the DHCP server

DHCP-Server → **Desktop** → **IP Configuration** → **Static**: IP 192.168.10.2, mask 255.255.255.0, gateway 192.168.10.1, DNS 192.168.10.2.

Screenshot 2
DHCP-Server static IP configuration

Step 3 — Configure the DHCP service

1. DHCP-Server → **Services** → **DHCP**; set Service **On**.
2. Edit the pool: name NPC-Pool, gateway 192.168.10.1, DNS 192.168.10.2, start IP 192.168.10.10, mask 255.255.255.0, max users 50.
3. Click **Save** (or Add if you create a new pool).

Screenshot 3
Services → DHCP with the pool saved and service On

Step 4 — Set PCs to DHCP

On each PC: **Desktop** → **IP Configuration** → **DHCP**. Wait for "DHCP request successful".

Screenshot 4

PC0 IP Configuration set to DHCP, showing the received address

4. Verification and Results

4.1 Check received configuration

```
C:\> ipconfig /all
```

PC	IP address	Mask	Gateway	DNS
PC0	192.168.10.10 (expected)	255.255.255.0	192.168.10.1	192.168.10.2
PC1	192.168.10.11 (expected)	255.255.255.0	192.168.10.1	192.168.10.2
PC2	192.168.10.12 (expected)	255.255.255.0	192.168.10.1	192.168.10.2

Replace expected values with what your PCs actually received.

Screenshot 5
ipconfig /all on each PC

4.2 Connectivity

```
C:\> ping 192.168.10.11
C:\> ping 192.168.10.12
C:\> ping 192.168.10.2
```

Screenshot 6
Successful pings between PCs and to the DHCP server

4.3 DHCP bindings

DHCP-Server → Services → DHCP → the address table lists leased IPs and MAC addresses.

Screenshot 7
DHCP lease/binding table on the server

5. Troubleshooting Notes

- PC shows 169.254.x.x (APIPA) → DHCP service is Off, pool not saved, or cable not connected.
- If the pool start IP overlaps 192.168.10.2, the server could be handed out — start the pool at .10.
- Click DHCP again or run `ipconfig /renew` to re-request.

6. Conclusion

A DHCP server with a static address was configured in Cisco Packet Tracer with an address pool, mask, default gateway and DNS server. All PCs obtained correct settings automatically and could ping each other, showing how DHCP removes manual addressing, reduces errors and makes network administration easier.